

Esercizi

Edoardo Longo

Svolgere i seguenti integrali:

1.

$$\int \frac{x^2 + 2}{x^2} dx$$

2.

$$\int \frac{x^2 + 2}{x^3} dx$$

3.

$$\int \frac{(x^2 - 1)^2}{x^2} dx$$

4.

$$\int \frac{x^2 - 4x^2 - \sqrt{x} - 1}{x} dx$$

5.

$$\int \frac{x \sin x + 1}{x} dx$$

6.

$$\int (e^x - \cos x) dx$$

7.

$$\int \frac{e^{3x} \cdot e^{2x}}{e^{4x}} dx$$

8.

$$\int e^{2x+4} \cdot e^{-2x-3} dx$$

9.

$$\int \left(\frac{2 + 4xe^x}{x} \right) dx$$

10.

$$\int (x - 1)^2 (x + 1) dx$$

11.

$$\int \frac{xe^x + e^{2x}}{e^x} dx$$

12.

$$\int (x^2 + 2 \sin x - e^x - 1) dx$$

13.

$$\int \frac{x^2 e^x + 1}{x^2} dx$$

14.

$$\int \left(x + \frac{1}{2x^2 + 2} \right) dx$$

15.

$$\int \left(\frac{e^{3x-1}}{e^{2x-1}} + e^x \cdot e^{2-x} \right) dx$$

16.

$$\int \frac{\sqrt{x} + \sqrt[3]{x}}{x} dx$$